



HTOC

HEALTHCARE THREAT OPERATIONS CENTER

I&W Batch Reporting Pipeline (Core + Expanded)

Version 1.0
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Table of Contents

Table of Contents	3
1. Purpose	5
2. Scope	5
3. Prerequisites	5
3.1 Runtime and Access	5
3.2 Data dependencies	6
4. Process Components	6
5. Required Execution Order	7
6. Core Path Behavior	8
6.1 I&W_Spreadsheet.py	8
6.2 I&W_Generator.py	8
7. Expanded Path Behavior	9
7.1 I&W_Document_expanded_spreadsheet.py	9
7.2 I&W_Document_expanded_generator.py	9
8. Batch Launcher Behavior	10
9. Run Success Criteria	10
10. Troubleshooting	11
11. Operator Decision Points	12
12. How to Run Manually	12
13. Related Documents	13

Table of Contents

Table of Contents	3	2
1. Purpose	4	2
2. Scope	4	2
3. Prerequisites	4	2
4. Process Components	5	2
5. Required Execution Order	6	2
6. Core Path Behavior	7	2
7. Expanded Path Behavior	8	2
8. Batch Launcher Behavior	9	2
9. Run Success Criteria	9	2
10. Troubleshooting	10	2
11. Operator Decision Points	11	2
12. How to Run Manually	11	2
13. Related Documents	12	2

Table of Contents 3

1. Purpose	5
2. Scope	5
3. Prerequisites	5
3.1 Runtime and Access	5
3.2 Data dependencies	6
4. Process Components	6
5. Required Execution Order	7
6. Core Path Behavior	8
6.1 I&W_Spreadsheet.py	8
6.2 I&W_Generator.py	8
7. Expanded Path Behavior	9
7.1 I&W_Document_expanded_spreadsheet.py	9
7.2 I&W_Document_expanded_generator.py	9

8. Batch Launcher Behavior	10
9. Run Success Criteria	10
10. Troubleshooting	11
11. Operator Decision Points	12
12. How to Run Manually	12
13. Related Documents	13

1. Purpose

This SOP documents how to operate the full I&W batch workflow that identifies qualifying indicators, builds spreadsheet queues, enriches indicators with VirusTotal and OTX context, and generates Word-ready I&W reporting artifacts for analyst review and distribution.

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2. Scope

This procedure is for operators maintaining daily I&W data production. It covers the core path and expanded path, including dependency checks, execution order, expected outputs, and actions to take when a run fails or returns no indicators.

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3. Prerequisites

3.1 Runtime and Access

Requirement	Notes
Python runtime on execution host	Batch files point to either 'Z:\HTOC\JA\Python313\python.exe' or '\\10.1.4.22\data\HTOC\Data_Analytics\Py\python.exe'
ThreatConnect SDK path	'Z:\HTOC\Data_Analytics\threatconnect' must be accessible

Requirement	Notes
ThreatConnect config file	`Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\I&W Report Processing Scripts\utils\config.json`
Access to I&W staging paths on `Z:`	Required for spreadsheet/report/log read-write operations
Task Scheduler access on F.R.E.D	Needed to verify and rerun scheduled tasks

Example dependency install command:

```
pip install pandas openpyxl requests python-docx urllib3 pytz
```

3.2 Data dependencies

OpDiv observation source files must exist for the last 2-3 days.

ThreatConnect API credentials must be valid and active.

For expanded runs, `indicators.csv` must exist (or be creatable) at:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Reported Indicators\indicators.csv](#)

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4. Process Components

Component	Path	Role
Core spreadsheet builder	`notebooks/I&W Reporting/Batch/I&W Spreadsheet/I&W_Spreadsheet.py`	Pulls candidate indicators and writes full spreadsheet for report generation

Component	Path	Role
Core report generator	`notebooks/I&W Reporting/Batch/I&W Generator/I&W_Generator.py`	Reads latest core spreadsheet, enriches indicators, creates Word reports
Expanded spreadsheet builder	`notebooks/I&W Reporting/Batch/I&W Expanded/I&W_Document_expanded_spreadsheet.py`	Builds expanded indicator spreadsheet and filters already-reported items
Expanded report generator	`notebooks/I&W Reporting/Batch/I&W Expanded/I&W_Document_expanded_generator.py`	Reads latest expanded spreadsheet, enriches, creates expanded report set, updates indicators list
Batch launchers	`run_iw_*.bat` files listed in header table	Install dependencies, run scripts, and write log output

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5. Required Execution Order

Run in this order for complete daily processing:

```
run_iw_spreadsheet.bat
run_iw_generator.bat
run_iw_expanded_spreadsheet.bat
```

run_iw_expanded_generator.bat

If step 1 or step 3 produces no spreadsheet output, do not run its corresponding generator step until source-data conditions are corrected.

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6. Core Path Behavior

6.1 I&W_Spreadsheet.py

Queries ThreatConnect [/v3/indicators](#) for HTOC org and selected type list.

Loads recent OpDiv observations from:

[Z:\HTOC\Data Analytics\Data\OpDiv Observations\htoc opdiv obs d{YYYYMMDD}.csv.](#)

Keeps indicators observed by multiple partners and filters out undesired tags (for example htoc_w1).

Fetches indicator attributes and excludes SOAR-authored attributes when present.

Writes output spreadsheet to:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Spreadsheet\I&W indicators full {YYYYMMDD}.xlsx.](#)

6.2 I&W_Generator.py

Loads the latest core spreadsheet (I&W_indicators_full_*.xlsx).

Enriches each indicator using ThreatConnect enrich (Shodan, VirusTotalV3) and OTX API.

Groups indicators by group_id when available.

Fills I&W Report Template.docx placeholders and table rows.

Writes reports to:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Generated Reports\{YYYY-MM-DD}\.](#)

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7. Expanded Path Behavior

7.1 I&W_Document_expanded_spreadsheet.py

Repeats indicator collection and partner-filter pipeline for expanded processing.

Loads prior reported indicators from:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Reported Indicators\indicators.csv.](#)

Removes already-reported indicators from current batch.

Writes expanded spreadsheet to:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Spreadsheet\Expanded\expanded indicators {YYYYMMDD}.xlsx.](#)

7.2 I&W_Document_expanded_generator.py

Loads most recent file from:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Spreadsheet\Expanded\.](#)

Enriches indicators with ThreatConnect and OTX.

Generates report documents in:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Expanded Reports\{YYYY-MM-DD}\.](#)

Appends processed indicators to:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\Reported Indicators\indicators.csv.](#)

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8. Batch Launcher Behavior

All four batch files perform operator-safe guardrails:

Verify Python executable path exists.

Verify target script path exists before execution.

Install required packages (`pip install ...`) before run.

Capture script output and write timestamped log files under:

[Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\logs\](#).

Return non-zero exit code on failure.

The hardened launchers (`run_iw_spreadsheet.bat` and `run_iw_expanded_spreadsheet.bat`) also support:

Local temp directory ([C:\Temp](#)) for pip reliability.

Optional wheelhouse fallback at [Z:\HTOC\JA\wheelhouse](#).

Additional success/failure summaries printed at end of run.

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9. Run Success Criteria

A successful full-cycle run should satisfy all checks:

Each batch launcher exits cleanly (Task Scheduler result 0x0 when scheduled).

Core spreadsheet exists for current date in [Spreadsheet\](#).

Core report folder for current date exists in [Generated Reports\](#).

Expanded spreadsheet exists for current date in [Spreadsheet\Expanded\](#).

Expanded report folder for current date exists in [Expanded Reports\](#).

Latest log files in [logs\](#) show success state and no unhandled tracebacks.

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10. Troubleshooting

Symptom	Likely Cause	Resolution
`Python executable not found` in batch output	Runtime path changed or inaccessible	Update launcher `PYTHON_EXE` to valid host path and retest interactively
`Failed to load configuration`	Missing/invalid `config.json` or no read access	Validate config file path and permissions; confirm keys (`access_id`, `secret_key`, `base_url`, `default_org`)
`No indicators retrieved` or `No data to export`	No qualifying indicators in date window	Confirm source observation files and ThreatConnect data freshness; rerun after upstream update
OTX/VirusTotal errors per indicator	External API throttling/network issue	Retry run; if persistent, keep core spreadsheet output and proceed with partial context note
No reports generated but spreadsheet exists	Template path issue or placeholder mismatch	Verify `I&W Report Template.docx` path and template integrity
Expanded run returns empty after filtering	All candidates already in `indicators.csv`	Expected behavior; validate with analyst before manual override

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11. Operator Decision Points

If core spreadsheet is empty, confirm whether this is an expected low-activity day before escalation.

If enrichment partially fails but critical outputs exist, decide whether to publish with reduced context or rerun.

If expanded filtering removes all indicators, confirm no urgent operational need exists before bypassing reported-indicator controls.

If any batch step fails, fix root cause and rerun that step before continuing to downstream steps.

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12. How to Run Manually

From Task Scheduler host (or equivalent execution host), run each launcher in order:

```
cmd /c "Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\I&W Report Processing Scripts\Spreadsheet_scripts\run_iw_spreadsheet.bat"
cmd /c "Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\I&W Report Processing Scripts\Spreadsheet_scripts\run_iw_generator.bat"
cmd /c "Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\I&W Report Processing Scripts\Expanded Scripts\run_iw_expanded_spreadsheet.bat"
cmd /c "Z:\HTOC\HTOC Reports\I&W Reports\5. I&W Staging\I&W Report Processing Scripts\Expanded Scripts\run_iw_expanded_generator.bat"
```

After each step, confirm:

Exit code is 0.

Expected output file/folder for that step exists.

Corresponding log file is written to the logs directory.

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13. Related Documents

SOP_Next_Obs_Daily_Batch.md

SOP_Daily_Reports_Consolidation.md

GitHub repository: [jaylinaskew-OIS/HTOC](https://github.com/jaylinaskew-OIS/HTOC)